```
hddatanode2
hddatanode3
```

Core-site.xml: Change the host name from *localhost* to *hdnamenode* in *core-site.xml* as shown below:

```
hduser@hdnamenode:~$vi /usr/local/hadoop/etc/hadoop/core-site.xml
<configuration>
<property>
<name>fs.default.name</name>
<value>hdfs://hdnamenode:9000</value>
</property>
<property>
<name>hadoop.tmp.dir</name>
<value>/usr/local/hadoop/etc/hadoop/tmp</value>
</property>
</configuration>
```

Hdfs-site.xml: Change the replication value to 4, as there are four nodes including Hadoopmaster, as shown below.

Create two directories for Hadoop storage:

```
hduser@hdnamenode:~$mkdir -p /usr/local/hadoop_store/hdfs/namenode
hduser@hdnamenode:~$mkdir -p /usr/local/hadoop_store/hdfs/datanode
hduser@hdnamenode:~$sudo chown hadoop:hadoop -R /usr/local/hadoop_store
hduser@hdnamenode:~$vi /usr/local/hadoop/etc/hadoop/hdfs-site.xml
<configuration>
<property>
<name>dfs.datanode.data.dir</name>
<value>file:///home/hduser/hdfs/datanode</value>
<description>DataNode directory</description>
</property>
<property>
<name>dfs.namenode.name.dir</name>
<value>file:///home/hduser/hdfs/namenode</value>
<description>NameNode directory for namespace and transaction logs storage.</description>
</property>
<property>
<name>dfs.replication</name>
<value>2</value>
</property>
 </configuration>
```

Mapred-site.xml: Change the hostname from *localhost* to *Hadoopmaster* with port 54311:

```
<configuration>
<property>
    <name>mapreduce.framework.name</name>

    <value>yarn</value>

</property>
 </configuration>
```

Note: In the configuration file (*/etc/hosts*), *masters* and *slaves* should be the same in *namenode* and all the *datanodes*.

Once the core files of Hadoop are configured, it's time for formatting and starting/stopping the HDFS file system via *namenode:*

```
$hadoop namenode -format
```

You can start a multi-node cluster in two steps:

i. Start HDFS daemons, i.e., the NameNode daemon on Hadoopmaster and DataNode daemon on all the slave nodes.

ii. Start MapReduce daemons, i.e., JobTracker on *hdnamenode* and TaskTracker on the *hddatanode1*, *hddatanode2* and *hddatanode3* nodes.

```
hduser@hdnamonode:~$start-dfs.sh # this is to check NameNode and DataNode running or not
hduser@hdnamenode1:~$jps #will list all the java processes running on master & slave nodes
hduser@hdnameno1:~$ start-mapred.sh
hduser@hdnamenode1:~$ jps # to check Job tracker and task tracker running or not
hduser@hdnamenode1:~$stop-dfs.sh # to stop the service
```

Web services

You can check the Web interface of the NameNode, DataNode, MapReduce and TaskTracker processes at the following URLs:

```
http://hdnamenode:50070/ – web UI of the NameNode daemon
http://hdnamenode1:50030/ – web UI of the JobTracker daemon
http://hdnamenode:50060/ – web UI of the TaskTracker daemon
```

END

By: Vinay Patkar and Avinash Bendigeri

Vinay Patkar works as a software development engineer at Dell India R&D Centre, Bengaluru, and has close to two years' experience in automation, Windows Server OS. He is interested in virtualisation and cloud computing technologies.

Avinash Bendigeri, too, works as a software development engineer at Dell R&D Centre, Bengaluru. He is interested in the automation and systems management domains.